Fluorescence hyperspectral imaging detects an early, FLS2-independent signature of *Pseudomonas syringae* challenge in *Arabidopsis* seedlings, but not of flg22 perception

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Abstract

Detecting plant infection before symptoms appear would help both field and controlled-environment agriculture. In 2016 we imaged *Arabidopsis* seedlings by fluorescence hyperspectral imaging (488 nm excitation, 500–800 nm emission) before and 1–48 h after flood treatment with a mock solution, the flagellin epitope flg22 or *Pseudomonas syringae*. Five genotypes were imaged: Col-0, two *fls2* receptor mutants and two *cml24* Ca^{2+}-sensor mutants. We reanalysed all 1,982 hand-drawn seedling spectra of the primary experiment, and 400 raw image cubes (396 GB). Recovering the camera metadata showed that exposure time changed between timepoints: the 2016 summaries' 21% fall in intensity after treatment disappears (to within 4%) once intensities are divided by exposure. Treated and control plates also differed *before* treatment, enough for a classifier to tell them apart (plate-level AUC 1.00), so we analysed only change from each plate's own baseline. On that basis, *P. syringae* broadened the red chlorophyll emission peak within 1 h (+1.41 nm; $q = 0.045$). By 2 h it had also raised emission below 640 nm, and change-from-baseline spectra separated infected from control plates perfectly at 1, 2 and 24 h (leave-plate-out AUC 1.00; exact permutation $p = 0.029$). The pathogen signature appeared in every genotype. A flg22 signal was weaker (AUC 0.89–1.00, $p \geq 0.10$). It was as strong in both *fls2* mutants, which cannot perceive flg22, as in Col-0, so in this experiment it reflects a plate-level effect rather than flg22 perception. We release the raw data, manifests, code and a reproducible pipeline as a FAIR resource, and document every data problem we found.

1 Introduction

Plants perceive bacteria through pattern-recognition receptors long before disease symptoms appear. The best-characterised receptor is FLS2, which binds the conserved flg22 epitope of bacterial flagellin [1–4]. flg22 induces defence-gene expression and resistance in wild-type plants but not in *fls2* mutants. *fls2* plants still respond to bacterial extracts through other patterns, and are more susceptible to *P. syringae* pv. tomato DC3000 sprayed on the leaf surface [5]. Pattern-triggered defence quickly lowers non-photochemical quenching [6]. *P. syringae* infection changes photosynthetic chlorophyll fluorescence parameters near the infection site before visible symptoms, within 3 h for an avirulent strain [7]. Chlorophyll fluorescence imaging is therefore a candidate for pre-symptomatic disease detection [8–11].

Steady-state chlorophyll fluorescence emission has two bands, near 685–690 nm and 730–740 nm. Because chlorophyll reabsorbs the red band, their ratio falls as chlorophyll content rises. The ratio

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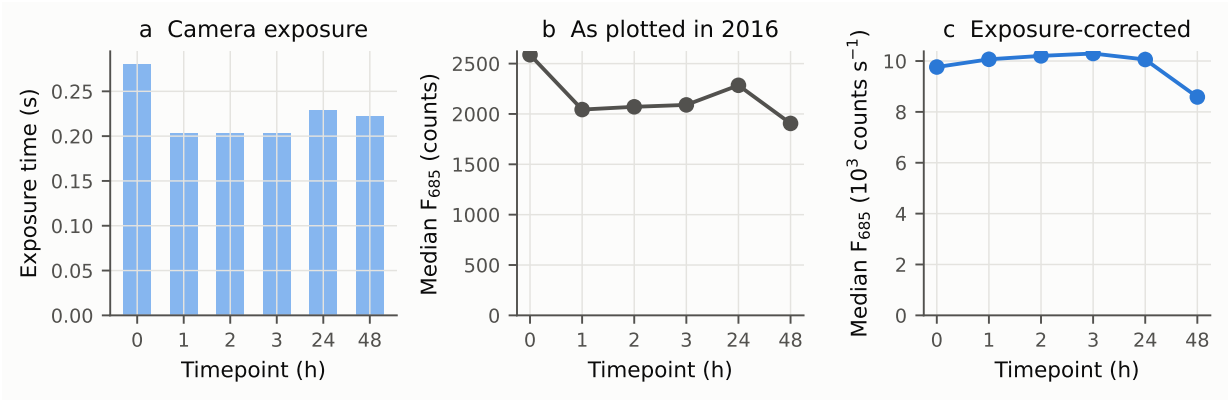


Figure 1: The 2016 intensity drop after treatment is an exposure artefact. (a) Exposure time per timepoint (plate mean, Control plates). (b) Median 683–687 nm emission of Control hand ROIs as plotted in 2016 and (c) divided by exposure. Data: `results/roi_spectra_A.parquet`, `metadata/cube_manifest.csv`.

also depends on leaf structure, optics and photosynthetic activity [12]. Fluorescence hyperspectral imaging records the whole emission spectrum at every pixel, so subtle shape changes can be mapped, not just one or two ratios.

CML24 (*TCH2*) encodes a calmodulin-like Ca^{2+} sensor whose transcript is induced by touch, darkness and other stimuli [13]. The *cml24-4* mutant is impaired in pathogen- and pattern-induced nitric-oxide production [14]. This makes CML24 a candidate link between early Ca^{2+} signalling and downstream changes visible in fluorescence.

Here we reanalyse a 2016 experiment designed to ask whether bacterial challenge and flg22 perception leave a hyperspectral fluorescence signature, and whether that signature depends on FLS2 or CML24. Its fls2 mutants supply the key control: an flg22 signature that reflects perception should be absent in them.

2 Results

2.1 Recovering the experiment from its raw files

We inventoried 7,752 files and parsed every file name into a design record, with no exceptions. That gave 216 sample cubes for the primary July experiment (set A), 168 for an independent November replicate (set B) and 5,842 hand-drawn ROI spectra (2,486 unique by content). The headers revealed that exposure changed between timepoints (Fig. 1a). The 2016 graphs plotted raw counts, so their drop in Control intensity from 2,588 to 2,044 counts after hour 0 (Fig. 1b) is largely a camera setting. Exposure-corrected Control intensities at 1–3 h lie within 4% of hour 0 (Fig. 1c). Hour-0 scans also saturated the 12-bit detector in up to 13% of a cube's seedling pixels (median 0.1%). We therefore based the primary analysis on exposure-independent shape features.

2.2 Plates differed before treatment

Hour 0 was imaged before treatment. Even so, flg22 plates already differed from Control plates in red-peak width (-1.29 SD; $q = 1.1 \times 10^{-4}$). A classifier separated them perfectly from spectra alone (plate-level AUC 1.00, $p = 0.029$; Fig. 4a). The same held, less strongly, for Pathogen plates (AUC 0.81). Plate-to-plate differences in seedling batch, placement or imaging geometry are thus as large as the effects we want to detect. All inference below uses change from each plate's own hour-0 baseline, with the plate as the experimental unit.

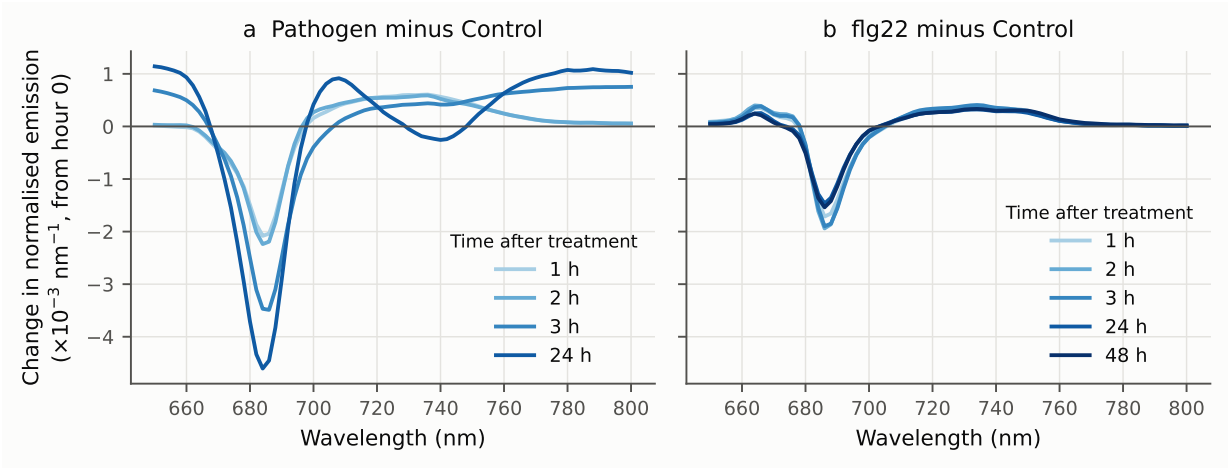


Figure 2: Treatment-induced change in emission-spectrum shape (Col-0, set A; area-normalised spectra, change from each plate's hour 0, treated minus Control).

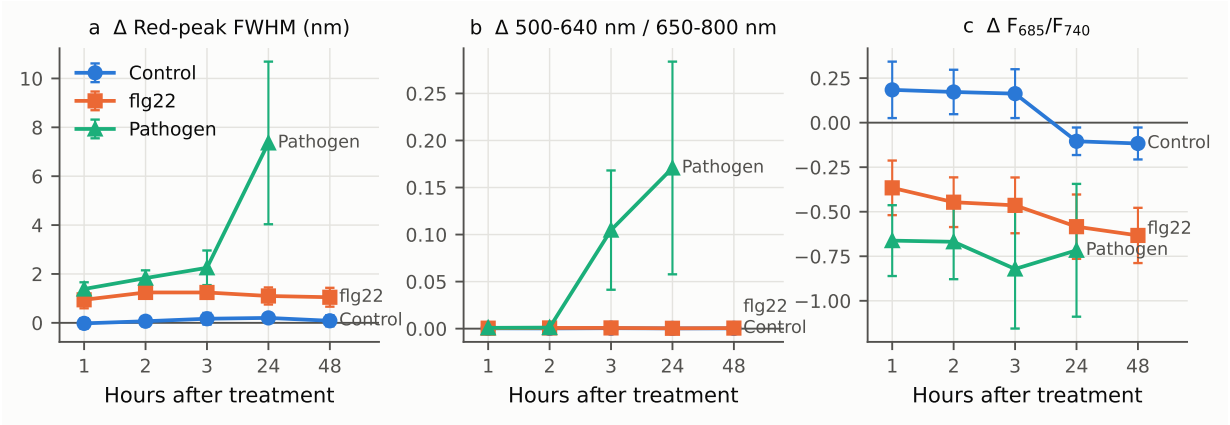


Figure 3: Time course of three emission-shape features (change from hour 0; mean ± s.e.m. over plates). Data: `tables/fig3_timecourse.csv`.

2.3 An early pathogen signature

Relative to baseline, *P. syringae* broadened the red emission peak of Col-0 seedlings by 1.41 nm at 1 h (2.07 SD, $q = 0.045$) and 1.77 nm at 2 h (2.39 SD, $q = 0.001$). It also raised the 500–640 nm fraction of emission at 2 h (2.26 SD, $q = 0.005$; Figs 2, 3). These were the only 3 of 270 baseline-adjusted tests significant at $q < 0.05$. The spectral change deepened through 24 h (Fig. 2a). A classifier on change-from-baseline spectra separated infected from control plates perfectly at 1, 2 and 24 h (AUC 1.00, 1.00, 1.00; exact $p = 0.029$ each, close to the smallest p the 70 plate relabellings allow; Fig. 4b). It failed at 3 h [PENDING: explain or flag the 3 h dip once cube-level data are in]. The 2016 vendor analysis of a single infected and a single control plate reported a broad autofluorescence component in infected plants after 24 h (FACTS V1). Our short-wavelength fraction measures that component across all plates.

2.4 The flg22 signal does not depend on FLS2

After baseline adjustment, flg22 plates were separable from Control plates at 1–3 h (AUC 0.89–1.00; Fig. 4b). With three plates per treatment left, however, no result beat $p = 0.10$ (20 relabellings). The genetics decide what this signal is. A model trained on Col-0 flg22-versus-Control shape features gave an average flg22 score difference of 0.21 in Col-0 (out-of-fold, 1–3 h), 0.36 in *fls2-17* and 0.20 in *fls2-101* (Fig. 5a). flg22 cannot act through FLS2 in the mutants, so this signal is a property of the plates (handling, timing or seedling batch), not of flg22 perception. The pathogen

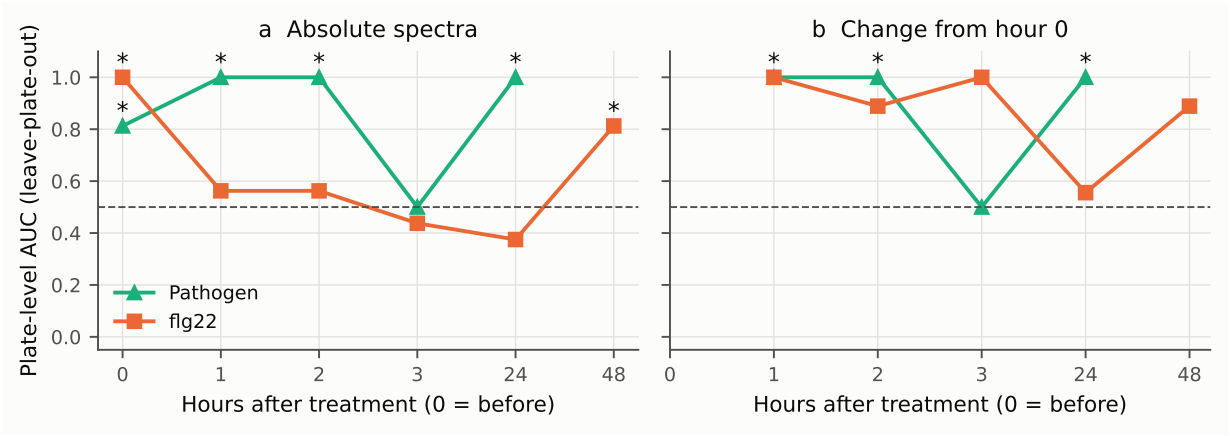


Figure 4: Plate-level classification of treatment from spectra, Col-0. (a) Absolute spectra; hour 0 is before treatment. (b) Change from hour 0. * exact permutation $p \leq 0.05$.

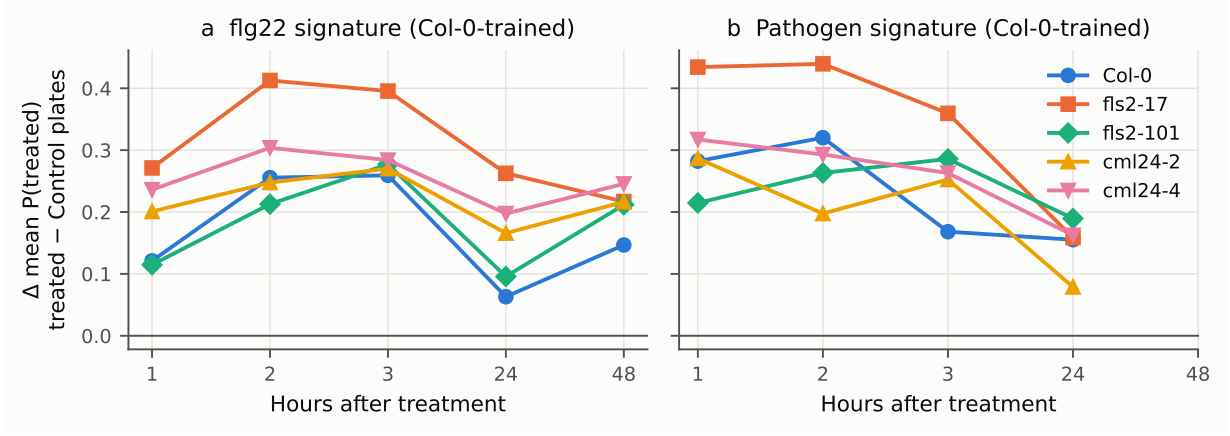


Figure 5: Genotype specificity of the signatures. Col-0-trained model applied to each genotype; difference in mean predicted probability between treated and Control plates.

signature was likewise present in all genotypes (Fig. 5b). This is expected, because bacteria are perceived through patterns other than flagellin [5].

2.5 The independent replicate

In the November replicate (set B; 405 nm excitation), hand ROIs exist for one plate per treatment, so treatment main effects cannot be separated from plate effects there. Within plates, however, flg22 changed the emission shape of Col-0 differently from *fls2-17* at 6 h (F_{685}/F_{740} interaction 1.78 SD, $q = 0.026$; 4 related features). This is the pattern expected of FLS2-dependent perception. It was not seen for *fls2-101*. [PENDING: Plate-replicated set-B analysis from the raw cubes (2–3 plates per treatment).]

2.6 Seedling-level analysis of the raw cubes

[PENDING: Segmentation of all cubes (Pathogen 48 h, exposure-corrected intensity, spatial heterogeneity) and pixel-level multivariate curve resolution [15]; label-transfer validation in Supplementary Note 1.]

3 Discussion

Three lessons from this reanalysis apply beyond these data. First, acquisition metadata must travel with hyperspectral intensities. A camera setting that changed between timepoints produced the most striking pattern in the 2016 summaries. Second, in plate-based seedling assays the plate, not the seedling, is the treated unit. Plates can differ before treatment by as much as treatments differ afterwards, so a baseline image and leave-plate-out validation are essential. Without them, a classifier here would have "detected" flg22 before it was applied. Third, mutants that cannot perceive the stimulus are the decisive control. Here they showed that an apparently promising flg22 signal was not perception.

What survives is a pathogen signature within 1–2 h of flood inoculation: a broader red emission peak and more short-wavelength emission. The timing is consistent with reports that chlorophyll fluorescence changes precede visible symptoms [7]. Because the signature appears in *fls2* mutants, it does not require flagellin perception. [PENDING: Whether CML24 modulates it awaits the seedling-level analysis; the hand-ROI data show no cml24 × treatment interaction at $q < 0.05$.] [PENDING: Limitations: pathovar/strain, inoculum and seedling age unrecorded on the drive; four plates per treatment; set-B excitation differs.]

4 Methods

4.1 Plant material and treatments

Arabidopsis thaliana seedlings of five genotypes were grown on square agar plates in five lanes per plate: Col-0 (lane A), *fls2-17* (B), *fls2-101* (C), *cml24-2* (D) and *cml24-4* (E). Lane assignments are taken from the 2016 analysis folders (FACTS R8). [PENDING: Allele origins, backgrounds and stock numbers.] *FLS2* encodes the flagellin receptor that perceives the flg22 epitope [1–4]. *CML24* encodes a calmodulin-like Ca^{2+} sensor [13] implicated in flg22-triggered nitric-oxide signalling [14]. Seedlings were grown on [PENDING: medium (protocol draft: ½ LS, 0.3% P), age, light, photoperiod and temperature]. Plates were treated with a mock solution, flg22 or *Pseudomonas syringae* (FACTS R5) [PENDING: pathovar/strain, inoculum density, flg22 concentration]. The 2016 protocol draft specifies flood inoculation after Ishiga et al. [16], with Silwet L-77 as the mock (FACTS P1–P3). Four plates per treatment were imaged in the July experiment (set A).

The independent November experiment (set B; FACTS R7, R9) used Col-0, *fls2-17* and *fls2-101* only, with two or three plates per treatment, three rows per plate and three seedlings per genotype per row. A pilot experiment (February–March 2016) measured rosette leaf area from images of eight genotypes, including *glr3.3* and *glr3.4* alleles (FACTS R10–R11). It is reported in the supplement only.

4.2 Fluorescence hyperspectral imaging

Plates were imaged in the Gilroy Laboratory (UW-Madison Botany) with a MacroPhor™ fluorescence hyperspectral line-scan imager (Middleton Spectral Vision; FACTS R1). It carried an Andor Zyla 5.5 sCMOS camera in 12-bit low-noise mode (FACTS R2). Set A used 488 nm excitation and 500–800 nm emission (FACTS R3). The camera recorded 324 bands from 400.3 to 808.7 nm at about 1.23 nm spacing. Set B used 405 nm excitation and 400–800 nm emission (FACTS R4), recorded as 406 bands from 400.2 to 800.0 nm, each row-scan acquired at two settings labelled `Laser50` and `Laser58` [PENDING: meaning]. Each plate was imaged as three row-scans, one row of seedlings each. Set A was imaged before treatment (`Hour0`, 11:20–12:26 on 22 July 2016) and at 1, 2, 3, 24 and 48 h after infection (FACTS R6). Set B was imaged at 0, 6, 24 and 48 h. Acquisition settings were read from every ENVI header (`metadata/cube_manifest.csv`).

4.3 Data provenance and quality

All 7,752 files on the 2016 drive were inventoried and hashed (`scripts/00_inventory.py`). Every file name was parsed without exception: 400 hyperspectral cubes (216 set-A samples, 168 set-B samples, 8 set-B bacteria-only scans, 6 set-B reference scans, one reference and one artefact scan in set A) and 5,842 hand-drawn region-of-interest (ROI) spectra (2,486 unique by content hash). We found the following problems and corrected for each where possible.

1. **Exposure time changed between timepoints.** It was 0.305 s at `Hour0`, 0.203 s at 1–3 h, 0.222–0.268 s at 24 h (varying within plates) and 0.222 s at 48 h. The 2016 summary graphs plotted uncorrected intensities, so their apparent drop between 0 and 1 h (about 1.5-fold) matches the exposure ratio $0.305/0.203 = 1.50$. We divide every intensity by its own exposure time, and our primary features are exposure-independent ratios (Section 4.4).

2. **`Hour0` plates were scanned "inverted"** [PENDING: meaning], except four rescans labelled "NI" made later the same day at the 1 h settings. `Hour0` is used only as a baseline and is analysed separately from the upright timepoints.

3. **Detector saturation.** At the `Hour0` exposure, up to [PENDING: N]% of seedling pixels reached the 12-bit ceiling (4095 counts). Saturated pixels are excluded from all spectra.

4. **Fixed scan order.** Within each timepoint plates were scanned Pathogen, then flg22, then Control, 20–40 min apart, so scan time is partly confounded with treatment. We report the drift of Control plates between consecutive hours as a bound on this effect.

5. **The plate is the experimental unit.** Treatments were applied per plate (four per treatment in set A), so treatment effects are tested at plate level. Genotype contrasts are within plates.

6. **Missing cells.** No hand ROIs were drawn for Pathogen plates at 48 h, although the cubes exist. Our automated segmentation covers them.

4.4 Spectral processing

Each raw cube was read once, sequentially, and hashed with SHA-256 (`scripts/02_harvest_cubes.py`). The per-band dark level was taken as the median of background pixels, which sit at about 108 counts. Seedlings were segmented on the dark-subtracted 685 nm image at 10 background standard deviations, then cleaned, and cotyledon and hypocotyl components were grouped (`src/pathofluor/segment.py`). Each segmented seedling was assigned its lane and plant number from the row layout, which was validated against the 2016 hand ROIs (Supplementary Note 1). Spectral features were computed from the mean unsaturated spectrum of each seedling after Savitzky–Golay smoothing (`src/pathofluor/features.py`):

- the red/far-red ratio F_{685}/F_{740} [12];
- the position and full width at half maximum of the red emission peak;
- the spectral centroid;
- the far-red fraction of 650–800 nm emission;
- the short-wavelength fraction (500–640 nm emission relative to 650–800 nm), which captures the broad autofluorescence reported by the 2016 vendor analysis;
- exposure-normalised intensities (counts s^{-1});
- per-pixel heterogeneity of the ratio and of the short-wavelength fraction.

4.5 Statistics and classification

[PENDING: Final model specification, written after the analysis is frozen.] Treatment effects were tested with linear mixed models with plate as a random effect, against time-matched controls, with false-discovery-rate control [17]. Classifiers were evaluated by leave-one-plate-out cross-validation, with a plate-level permutation null.

4.6 Data and code availability

Code (MIT licence), derived data, the 2016 hand-ROI spectra and all manifests are in the repository. Raw cubes are deposited on Zenodo [PENDING: DOIs]. The metadata follow MIAPPE where applicable [18] and the FAIR principles [19].

Acknowledgements

[PENDING: Acknowledgements and funding (FACTS P10).]

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